Proposal

Due date: February 11th, 2026 at 11:59pm

For every checkpoint, it is important to mention at the end what you explicitly used generative AI for. If you did not use generative AI, then state you did not use it.

This checkpoint requires students to specify the details of their chosen project.

An example of a proposal, as well as other files, can be found here.

Submission Requirements

Upload:

- a maximum 2-page PDF,
- the file used to render the pdf,
- and an accompanying BibTeX file.
- **Project Topic**
  - State the topic idea you are pursuing (must already be approved by the instructor).
- **Simulation vs. Dataset**
  - Indicate whether your project will use pure simulation or a dataset.
  - If using a dataset:
    - Provide the source (datasets must come from published research articles).
    - Suggested repositories: Data Dryad, Zenodo.
    - If you cannot decide, the instructor may assign a dataset, which you must use.
    - Attach the dataset as a .csv file.
- **Project Details**
  - Expand on the requirements outlined in your chosen topic description.
  - For original topics, include the minimum expectations previously given by the instructor.
- **User Inputs (Shiny Components)**
  - List at least five interactive inputs you plan to include in your R Shiny app.

    - Examples: simulation sample size, random seed, plot color, option to download the simulated dataset as .csv, etc.
    - Creativity is encouraged. More inputs are welcome!
- **References**
    - Include as many references as possible.
    - Since you will be using R Shiny, begin with citing R and Shiny.
    - If you know in advance which R packages you'll use, include their citations (`citation("packageName")` in `R`).
    - If the reference is NOT a textbook, then it MUST include a DOI link. If no DOI exists, provide a valid URL. If neither exists, the source cannot be used.
    - The format of the citations do not matter, but it should be consistent amongst citations.

No more wasting your time to match the exact citation styles. Since you are expected to use BibTeX references, you'll need to write the proposal in LaTeX or anything else that supports BibTeX. [BibTeX works well with Rmarkdown](#) (and Quarto), and can easily be used.

You do not need to permanently commit to the details in the proposal; the example proposal and the finished product are actually different. However, it's good to write some ideas down so you have a direction to work towards.

Optional Responsibilities Document

Groups may add to their proposal a section detailing responsibilities of each group member for future checkpoints and designate a team leader. This section will be reviewed at the end when checking git history, and does not count towards the 2-page limit.

This section is optional but encouraged to ensure certain members don't over or under contribute.

Making Scheme

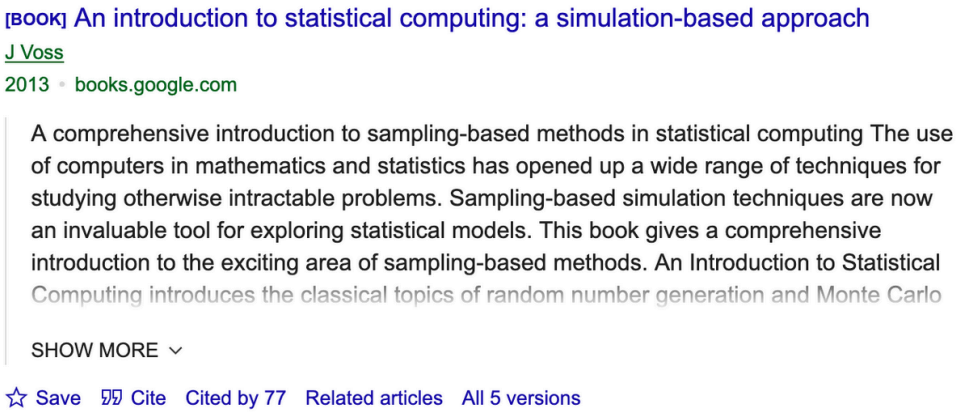
Assignments will directly be marked on the GitHub repository. If there are any missing files, then the assignment would be considered incomplete and will get a 0.

- 70% of marks are provided for proper completion.
- 20% of marks are awarded for correct BibTeX citation.
- 10% of marks are awarded for a statement regarding generative AI usage (or lack thereof.)

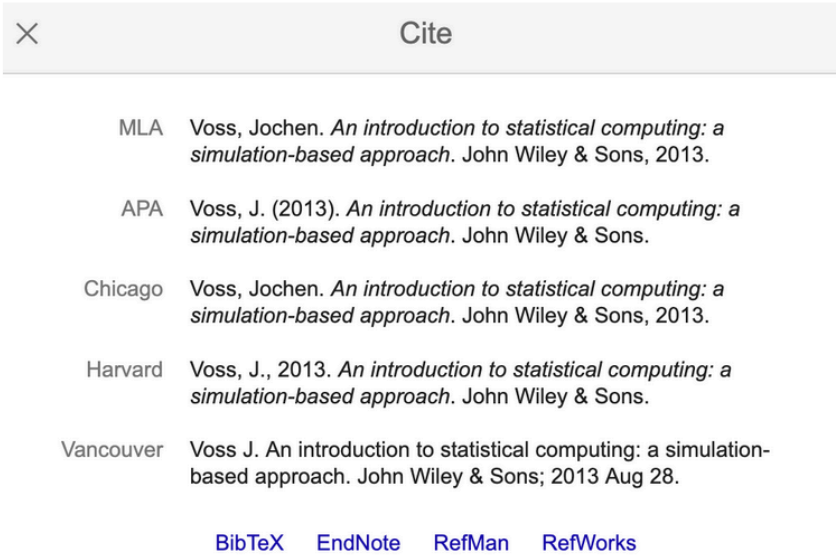
BibTeX Guide

From my personal experience, Generative AI tends to hallucinate when it comes to citations. This is how I do them easily.

1. Go to Google Scholar and search for the reference I want to use.



2. Press "Cite" (at the end, to the left of "Cited by XX").



3. Click "BibTeX" (see at the bottom, to the left of EndNote).

```
\@book{voss2013introduction,
  title={An introduction to statistical computing: a simulation-based approach},
  author={Voss, Jochen},
  year={2013},
  publisher={John Wiley \\& Sons}
}
```

4. Copy this output and paste it into the `.bib` file.

If you ever want to reference the citation in a sentence, we tend to use `@key`. In the above example, the key in that case would be voss2013introduction, so one would use `@voss2013introduction`.

In my example, I did not reference any citations. By default, citations only are referenced if you use then; to ensure that all are listed regardless of reference in the main text, you'll need to do something similar to `\nocite{*}`; see the example .tex file here.

Alternative Method: Zotero

Nowadays, a lot of researchers use Zotero. While this is a powerful tool, my personal issue is I find this a bit finicky when dealing with articles that do not have a DOI link. Nonetheless, it is also a useful asset to have.